Zhan Ling

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RESEARCH INTEREST

I am an explorer at heart, dedicated to pushing the frontiers of artificial intelligence. My long-term vision is to develop open-world, self-improving intelligent foundation models that drive us toward super-intelligence and expand the boundaries of human knowledge.

My career is defined by a relentless pursuit of diverse and challenging domains. During my Ph.D., I ventured broadly across imitation learning, reinforcement learning, robotics, 3D representation learning, and LLM reasoning. After that, I continued this exploration by driving LLM agent post-training at the cutting edge. I am continually exploring new technical frontiers, including training long-context LLMs via reinforcement learning and building omni real-time agents.

EDUCATION

University of California San Diego, La Jolla, California, USA 2019.9–2024.12
Doctor of Philosophy, Computer Science, Advisor: Prof. Hao Su
Master of Science, Computer Science

Tsinghua University, Beijing, China 2015.8–2019.7
Bachelor of Engineering, Institute for Interdisciplinary Information Sciences (IIIS)
Yao Class (founded by Turing Award Laureate, Prof. Andrew Chi-Chih Yao)

PUBLICATIONS AND PREPRINTS

(*, **, *** indicate equal contribution)

1. Seed2.0 Model Card: Towards Intelligence Frontier for Real-World Complexity. ByteDance Seed. Preprint.
2. Natural Language Actor-Critic: Scalable Off-Policy Learning in Language Space. Joey Hong, Kang Liu, **Zhan Ling**, Jiecao Chen, Sergey Levine. *Neural Information Processing Systems (NeurIPS) 2026.*
3. Scaling Long-Horizon LLM Agent via Context-Folding. Weiwei Sun, Miao Lu, **Zhan Ling**, Kang Liu, Xuesong Yao, Yiming Yang, Jiecao Chen. *International Conference on Machine Learning (ICML) 2026.*
4. Scaling LLM Multi-turn RL with End-to-end Summarization-based Context Management. Miao Lu, Weiwei Sun, Weihua Du, **Zhan Ling**, Xuesong Yao, Kang Liu, Jiecao Chen. *Annual Meeting of the Association for Computational Linguistics (ACL) 2026, Main.*
5. Seed1.8 Model Card: Towards Generalized Real-World Agency. ByteDance Seed. Preprint.

6. Generalizable End-to-End Tool-Use RL with Synthetic CodeGym. Weihua Du, Hailei Gong, **Zhan Ling**, Kang Liu, Lingfeng Shen, Xuesong Yao, Yufei Xu, Dingyuan Shi, Yiming Yang, Jiecao Chen. *International Conference on Learning Representations (ICLR) 2026*.

7. ORIC: Benchmarking Object Recognition in Incongruous Context for Large Vision-Language Models. Zhaoyang Li*, **Zhan Ling***, Yuchen Zhou, Litian Gong, Erdem Bıyık, Hao Su. *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR) 2026*.

8. MIR-Bench: Benchmarking LLM's Long-Context Intelligence via Many-Shot In-Context Inductive Reasoning. Kai Yan, **Zhan Ling**, Kang Liu, Yifan Yang, Ting-Han Fan, Lingfeng Shen, Zhengyin Du, Jiecao Chen. *Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track 2025*.

9. LongReason: A Synthetic Long-Context Reasoning Benchmark via Context Expansion. **Zhan Ling**, Kang Liu, Kai Yan, Yifan Yang, Weijian Lin, Ting-Han Fan, Lingfeng Shen, Zhengyin Du, Jiecao Chen. Preprint.

10. ActPlan-1K: Benchmarking the Procedural Planning Ability of Visual Language Models in Household Activities. Ying Su, **Zhan Ling**, Haochen Shi, Cheng Jiayang, Yauwai Yim, Yangqiu Song. *Empirical Methods in Natural Language Processing (EMNLP) 2024 Main*.

11. Unleashing the Creative Mind: Language Model As Hierarchical Policy For Improved Exploration on Challenging Problem Solving. **Zhan Ling**, Yunhao Fang, Xuanlin Li, Tongzhou Mu, Mingu Lee, Reza Pourreza, Roland Memisevic, Hao Su. Preprint.

12. DreamFuser: Value-guided Diffusion Policy for Offline Reinforcement Learning. Kairong Luo, Caiwei Xiao, Zhiao Huang, **Zhan Ling**, Yunhao Fang, Hao Su. Preprint.

13. Deductive Verification of Chain-of-Thought Reasoning. **Zhan Ling***, Yunhao Fang*, Xuanlin Li, Zhiao Huang, Mingu Lee, Roland Memisevic, Hao Su. *Neural Information Processing Systems (NeurIPS) 2023*.

14. On the Efficacy of 3D Point Cloud Reinforcement Learning. **Zhan Ling***, Yunchao Yao*, Xuanlin Li, Hao Su. Preprint.

15. Distilling Large Vision-Language Model with Out-of-Distribution Generalizability. Xuanlin Li*, Yunhao Fang*, Minghua Liu, **Zhan Ling**, Zhuowen Tu, Hao Su. *IEEE / CVF International Conference on Computer Vision (ICCV) 2023*.

16. Reparameterized Policy Learning for Multimodal Trajectory Optimization. Zhiao Huang, Litian Liang, **Zhan Ling**, Xuanlin Li, Chuang Gan, Hao Su. *Deep Reinforcement Learning Workshop, Neural Information Processing Systems (NeurIPS) 2022*; *International Conference on Machine Learning (ICML) 2023, **Oral***.

17. PartSLIP: Low-Shot Part Segmentation for 3D Point Clouds via Pretrained Image-Language Models. Minghua Liu, Yinhao Zhu, Hong Cai, Shizhong Han, **Zhan Ling**, Fatih Porikli, Hao Su. *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR) 2023*.

18. ManiSkill2: A Unified Benchmark for Generalizable Manipulation Skills. Jiayuan Gu*, Fanbo Xiang*, Xuanlin Li**, **Zhan Ling****, Xiqiang Liu**, Tongzhou Mu**, Yihe Tang**, Stone Tao**, Xinyue Wei**, Yunchao Yao**, Xiaodi Yuan, Pengwei Xie, Zhiao Huang, Rui Chen, Hao Su. *International Conference on Learning Representations (ICLR) 2023*.

19. Frame Mining: a Free Lunch for Learning Robotic Manipulation from 3D Point Clouds. Minghua Liu*, Xuanlin Li*, **Zhan Ling***, Yangyan Li, Hao Su. *Conference on Robot Learning (CoRL) 2022.*

20. Improving policy optimization with generalist-specialist learning. Zhiwei Jia, Xuanlin Li, **Zhan Ling**, Shuang Liu, Yiran Wu, Hao Su. *International Conference on Machine Learning (ICML) 2022.*

21. Approximate Convex Decomposition for 3D Meshes with Collision-Aware Concavity and Tree Search. Xinyue Wei*, Minghua Liu*, **Zhan Ling**, Hao Su. *ACM Transactions on Graphics (Proceedings of SIGGRAPH) 2022.*

22. Close the Visual Domain Gap by Physics-Grounded Active Stereovision Depth Sensor Simulation. Xiaoshuai Zhang*, Rui Chen*, Ang Li**, Fanbo Xiang**, Yuzhe Qin**, Jiayuan Gu**, **Zhan Ling****, Minghua Liu**, Peiyu Zeng**, Songfang Han***, Zhiao Huang***, Tongzhou Mu***, Jing Xu, Hao Su. *IEEE Transactions on Robotics (T-RO) 2023.*

23. ManiSkill: Generalizable Manipulation Skill Benchmark with Large-Scale Demonstrations. Tongzhou Mu*, **Zhan Ling***, Fanbo Xiang*, Derek Yang*, Xuanlin Li*, Stone Tao, Zhiao Huang, Zhiwei Jia, Hao Su. *Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track 2021.*

24. State Alignment-based Imitation Learning. Fangchen Liu, **Zhan Ling**, Tongzhou Mu, Hao Su. *International Conference on Learning Representations (ICLR) 2020.*

WORK EXPERIENCE

ByteDance Inc., San Jose 2024.12-Present
Seed-LLM
Research Scientist

End-to-End Multi-Turn Function-Call Training with Reinforcement Learning

- We develop a robust synthetic-data generation pipeline to enhance end-to-end function-call performance.
- We build an end-to-end function-call training pipeline for ByteDance Seed, achieving significant improvements over Seed-1.5 and Seed-1.6. The pipeline is deployed in production through Doubao Mobile and Doubao Her (Doubao Voice Agent).

GeneralAgent Task Construction

- We contribute agent-training data for Seed-2.1 by constructing long-horizon strategic game environments designed to enhance agent performance on Vending-Bench.

Generative Reward Modeling for Music Generation

- We develop the first reinforcement learning pipeline for speech-to-text (S2T) reasoning models, enabling generative reward modeling for high-fidelity music generation.
- We construct synthetic reinforcement learning data to strengthen general music understanding.

Full-Duplex Omni Real-Time Agent Post-Training (In Progress)

- We audit, curate, and refine existing supervised fine-tuning (SFT) data for the Omni real-time agent, improving dataset quality and consistency for full-duplex post-training.

Multi-Agent LLM Reinforcement Learning (In Progress)

- We develop a pipeline for synthesizing multi-agent reinforcement learning tasks.
- We design multi-agent reinforcement learning algorithms for personalized LLM agents.

ByteDance Inc., San Jose 2024.6-2024.12
Seed-LLM
Student Researcher, Host: Jiecao Chen

Building a Synthetic Benchmark and Dataset for Long-Context Reasoning

- We develop a novel algorithm to generate high-quality, long-context reasoning question-and-answer pairs.
- We create a synthetic long-context reasoning benchmark and evaluate both open- and closed-source models. This work resulted in a preprint and a patent application.

Qualcomm AI Research, San Diego 2023.6-2023.9
System 2 Team
Interim Engineering Intern, Host: Mingu Lee, Reza Pourreza, Roland Memisevic

Unleashing the Creative Mind: Language Model As Hierarchical Policy For Improved Exploration on Challenging Problem Solving

- Existing LLMs often have limited exploration capabilities in reasoning tasks, even when combined with sampling or tree search methods.
- Our key observation is that LLMs struggle to explore the high-level tactic space effectively.
- We propose an improved exploration strategy by using LLMs as a hierarchical policy, where the high-level policy sets the direction and the low-level policy handles the detailed reasoning steps.
- Details can be found in the paper [11].

Qualcomm AI Research, Remote 2023.3-2023.6
System 2 Team
Support Engineering, Host: Mingu Lee, Roland Memisevic
Deductive Verification of Chain-of-Thought Reasoning

- LLMs can produce unreliable reasoning chains containing logical errors and hallucinations.
- We propose a verifiable reasoning format called "Natural Program".
- We use LLMs for step-by-step self-verification to filter invalid reasoning chains and improve reasoning reliability.
- Details can be found in the paper [13].

Qualcomm AI Research, Remote 2022.6-2023.3
System 2 Team
Interim Engineering Intern, Host: Mingu Lee, Roland Memisevic
Learning Code Execution with Language Model

- We investigate whether LLMs can execute code as a Turing machine by training models on Python execution traces.
- We develop a random code generator supporting loops, conditionals, and function calls.
- We modify the interpreter to collect execution traces from generated code for model training.
- We train and evaluate LLMs on code-execution tasks, achieving high accuracy and strong generalization to unseen programs.
- This work resulted in a U.S. patent application.

X, the moonshot factory, Remote 2020.6-2020.9
Mineral Project
AI Resident, Host: Lianghao Li, Kangkang Wang
Lane Detection for Agricultural Imagery

- We develop a lane-detection algorithm based on the Hough transform in Python and accelerate it with Cython.

Improving Weed Detection with Lane Detection

- We create a weed detection model based on EfficientDet.
- We integrate the lane detection algorithm with the weed detection model to enhance overall weed detection performance.

University of California San Diego, San Diego 2019.9-2024.12
Su Lab
Graduate Research Assistant, Advisor: Hao Su
Embodied AI

- We co-develop the ManiSkill benchmarks for evaluating the generalization of robotic manipulation skills.
- We develop reinforcement-learning and imitation-learning algorithms for robotic manipulation tasks.

Computer Vision

- We develop 3D perception algorithms, including convex decomposition of 3D meshes and 3D part segmentation.

- We integrate 3D representation learning with reinforcement learning for robotic manipulation tasks.

Reasoning

- We develop algorithms to enhance reasoning abilities using LLMs.

AWARDS AND SERVICES

Conference Reviewer: CVPR 2022-2026; ICCV 2021, 2023, 2025; ECCV 2022, 2024, 2026; NeurIPS 2023-2025; ICLR 2024-2026; ICML 2024-2026; ACCV 2024; ICRA 2025; AISTATS 2025; ACL RR 2026.
Journal Reviewer: T-RO; RA-L.
Program Committee: AAAI 2025-2026.
Awards:

Outstanding Freshman Scholarship, Tsinghua University, 2015

First Prize (Gold Medal) in National Olympiad in Informatics (NOI), China, 2014

Workshop Reviewer:

Generalizable Policy Learning in the Physical World, ICLR 2022.

Interdisciplinary Exploration of Generalizable Manipulation Policy Learning: Paradigms and Debates, RSS 2023.

Challenge Organizer or Contributor:

SAPIEN ManiSkill Challenge 2021

SAPIEN ManiSkill Challenge 2022

TECHNICAL SKILLS

Programming Languages: Python, C, C++.
Deep Learning: PyTorch, TensorFlow, JAX, PyTorch CUDA extensions.
Large Models: Pre-training data processing, distributed training, fine-tuning, reinforcement learning, quantization.
Parallel Computing: MPI, OpenMP, CUDA, AVX.
Simulation: SAPIEN, MuJoCo.
Additional Skills: pybind11, ROS.

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